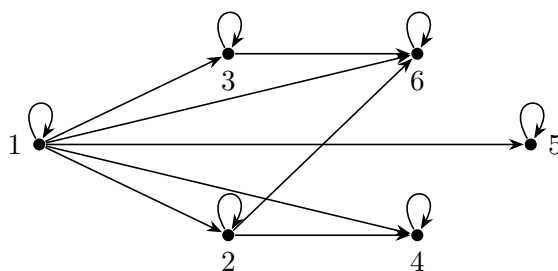


11 Relations

11.0 Relations

2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Write out the relation R that expresses $|$ (divides) on A . Then illustrate it with a diagram.

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$



6. Congruence modulo 5 is a relation on the set $A = \mathbb{Z}$. In this relation xRy means $x \equiv y \pmod{5}$. Write out the set R in set-builder notation.

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : 5 \mid (x - y)\}.$$

8. Let $A = \{1, 2, 3, 4, 5, 6\}$. Observe that $\emptyset \subseteq A \times A$, so $R = \emptyset$ is a relation on A . Draw a diagram for this relation.

Since $R = \emptyset$, no pair (x, y) belongs to R , so the diagram has a node for each element of A and no arrows at all.



10. Consider the subset $R = (\mathbb{R} \times \mathbb{R}) - \{(x, x) : x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$. What familiar relation on $\mathbb{R}$ is this? Explain.

The relation is *not equal* ($\neq$). The set contains every pair (x, y) of real numbers. Removing the pair (x, x) leaves exactly the pairs where x and y differ.

11.1 Properties of Relations

2. Consider the relation $R = \{(a, b), (a, c), (c, c), (b, b), (c, b), (b, c)\}$ on the set $A = \{a, b, c\}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is not reflexive because $(a, a) \notin R$.

This is not symmetric because $(a, b) \in R$ but $(b, a) \notin R$.

This is transitive: checking every chain xRy, yRz of pairs in R confirms that $(x, z) \in R$ in each case.

4. Let $A = \{a, b, c, d\}$. Suppose R is the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (a, c), (c, a), (a, d), (d, a), (b, c), (c, b), (b, d), (d, b), (c, d), (d, c)\}.$$

Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is reflexive because xRx holds for every $x \in A$.

This is symmetric because xRy and yRx both hold for every $x, y \in A$.

This is transitive because xRz holds for every $x, z \in A$, regardless of y .

6. Consider the relation $R = \{(x, x) : x \in \mathbb{Z}\}$ on $\mathbb{Z}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Here xRy if and only if $x = y$, so R is the equality relation $=$ on $\mathbb{Z}$.

This is reflexive because $x = x$ for every $x \in \mathbb{Z}$.

This is symmetric because $x = y$ implies $y = x$.

This is transitive because $x = y$ and $y = z$ imply $x = z$.

8. Define a relation on $\mathbb{Z}$ as xRy if $|x - y| < 1$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Since $x - y$ is an integer and the only integer with absolute value less than 1 is 0, we have xRy if and only if $x - y = 0$, that is, $x = y$. So R is the equality relation $=$ on $\mathbb{Z}$.

This is reflexive because $x = x$ for every $x \in \mathbb{Z}$.

This is symmetric because $x = y$ implies $y = x$.

This is transitive because $x = y$ and $y = z$ imply $x = z$.

10. Suppose $A \neq \emptyset$. Since $\emptyset \subseteq A \times A$, the set $R = \emptyset$ is a relation on A . Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is not reflexive. Pick any $x \in A$, then $(x, x) \notin R$ because $R = \emptyset$.

This is symmetric because it is impossible to find a $(x, y) \in R$ for which $(y, x) \notin R$.

This is transitive because $xRy \wedge yRz$ implies xRz for every $x, y, z \in A$.

12. Prove that the relation $|$ (divides) on the set $\mathbb{Z}$ is reflexive and transitive. (Use Example 11.8 as a guide if you are unsure of how to proceed.)

Proof. First we will show that $|$ (divides) is reflexive. Take any integer $x \in \mathbb{Z}$. Since $x = x \cdot 1$ and $1 \in \mathbb{Z}$, the definition of divisibility gives $x | x$. Thus $|$ is reflexive.

Next, we will show that $|$ is transitive. For this, we must show that if $x | y$ and $y | z$ then $x | z$. Suppose $x | y$ and $y | z$. This means $y = xm$ and $z = yn$ for some integers m and n . Substituting xm for y we get $z = (xm)n = x(mn)$, and by definition of divisibility, we see $x | z$.

The past two paragraphs show that $|$ is reflexive and transitive on $\mathbb{Z}$. □

14. Suppose R is a symmetric and transitive relation on a set A , and there is an element $a \in A$ for which aRx for every $x \in A$. Prove that R is reflexive.

Proof. Let R be a transitive and symmetric relation in A . And assume there is an element $a \in A$ such that aRx for every $x \in A$. We want to prove that xRx for every $x \in A$.

Take any $x \in A$. Since aRx for every $x \in A$, we have aRx . Because R is symmetric, we have xRa . So we have both xRa and aRx . Because R is transitive, xRa and aRx imply xRx . Therefore, R is transitive, symmetric, and reflexive. □

16. Define a relation R on $\mathbb{Z}$ by declaring that xRy if and only if $x^2 \equiv y^2 \pmod{4}$. Prove that R is reflexive, symmetric and transitive.

Proof. Assume R is a relation on $\mathbb{Z}$ such as xRy if and only if $x^2 \equiv y^2 \pmod{4}$. We want to prove that R is reflexive, symmetric, and transitive.

First, we will show that R is reflexive. Take any integer $x \in \mathbb{Z}$, and observe that $4 \mid 0$, so $4 \mid x^2 - x^2$. Thus, by definition, $x^2 \equiv x^2 \pmod{4}$. This shows R is reflexive.

Second, we will show that R is symmetric. Suppose xRy for some $x, y \in \mathbb{Z}$. So $4 \mid (x^2 - y^2)$. This means $(x^2 - y^2) = 4m$ for some $m \in \mathbb{Z}$. Then $y^2 - x^2 = 4(-m)$, so $4 \mid (y^2 - x^2)$. Thus by definition, $y^2 \equiv x^2 \pmod{4}$. Therefore R is symmetric.

Finally, we will show that R is transitive. Suppose xRy and yRz for $x, y, z \in \mathbb{Z}$. So $x^2 - y^2 = 4k$ for some integer k and $y^2 - z^2 = 4l$ for some integer l . Adding these two equations, we obtain $x^2 - z^2 = 4(k + l)$, so $4 \mid (x^2 - z^2)$. Hence $x^2 \equiv z^2 \pmod{4}$. Therefore R is transitive.

The past three paragraphs have shown that R is reflexive, symmetric and transitive. $\square$

11.2 Equivalence Relations

2. Let $A = \{a, b, c, d, e\}$. Suppose R is an equivalence relation on A . Suppose R has two equivalence classes. Also aRd , bRc and eRd . Write out R as a set.

Since aRd and eRd , the elements a , d and e lie in one equivalence class, and since bRc , the elements b and c lie in another. These are the two classes: $[a] = [d] = [e] = \{a, d, e\}$ and $[b] = [c] = \{b, c\}$. Each class X contributes the pairs $X \times X$ to R , so

$$R = \{(a, a), (d, d), (e, e), (a, d), (d, a), (a, e), (e, a), (d, e), (e, d), (b, b), (c, c), (b, c), (c, b)\}.$$

4. Let $A = \{a, b, c, d, e\}$. Suppose R is an equivalence relation on A . Suppose also that aRd and bRc , eRa and cRe . How many equivalence classes does R have?

R has one equivalence class. Since aRd and eRa , the elements a , d and e lie in one class, and since bRc and cRe , the elements b , c and e lie in one class. As e belongs to both, all five elements share a single class: $[a] = [b] = [c] = [d] = [e] = A$.

6. There are five different equivalence relations on the set $A = \{a, b, c\}$. Describe them all. Diagrams will suffice.

$$\begin{aligned} R_1 &= \{(a, a), (b, b), (c, c)\} \\ R_2 &= \{(a, a), (b, b), (c, c), (a, b), (b, a)\} \end{aligned}$$

$$\begin{aligned}
R_3 &= \{(a, a), (b, b), (c, c), (a, c), (c, a)\} \\
R_4 &= \{(a, a), (b, b), (c, c), (b, c), (c, b)\} \\
R_5 &= \{(a, a), (b, b), (c, c), (a, b), (b, a), (a, c), (c, a), (b, c), (c, b)\}.
\end{aligned}$$

8. Define a relation R on $\mathbb{Z}$ as xRy if and only if $x^2 + y^2$ is even. Prove R is an equivalence relation. Describe its equivalence classes.

Proof. By definition, xRy means $x^2 + y^2$ is even. We want to prove R is an equivalence relation, that is, R is reflexive, symmetric, and transitive.

First, we will show R is reflexive. Take any $x \in \mathbb{Z}$. Then $x^2 + x^2 = 2(x^2)$ is even, so xRx . Thus R is reflexive.

Second, we will show R is symmetric. Assume xRy for some $x, y \in \mathbb{Z}$. Since xRy , we know $x^2 + y^2$ is even, so $x^2 + y^2 = 2k$ for some integer k . From the commutative property of addition $x^2 + y^2 = y^2 + x^2$. So $y^2 + x^2 = 2k$. From this it follows that $y^2 + x^2$ is even, so yRx . Thus R is symmetric.

Last, we will show R is transitive. Assume that xRy and yRz for some $x, y, z \in \mathbb{Z}$. Since xRy and yRz , it follows $x^2 + y^2$ is even and $y^2 + z^2$ is even, so $x^2 + y^2 = 2m$ and $y^2 + z^2 = 2n$ for some $m, n \in \mathbb{Z}$. Adding these equations we get,

$$\begin{aligned}
(x^2 + y^2) + (y^2 + z^2) &= 2m + 2n \\
x^2 + y^2 + y^2 + z^2 &= 2m + 2n \\
x^2 + 2y^2 + z^2 &= 2m + 2n \\
x^2 + z^2 &= 2m + 2n - 2y^2 \\
x^2 + z^2 &= 2(m + n - y^2)
\end{aligned}$$

So $x^2 + z^2$ is even, so xRz . Thus R is transitive.

The three paragraphs have shown that R is reflexive, symmetric, and transitive. Therefore R is an equivalence relation. $\square$

10. Suppose R and S are two equivalence relations on a set A . Prove that $R \cap S$ is also an equivalence relation. (For an example of this, look at Figure 11.2. Observe that for the equivalence relations R_2, R_3 and R_4 , we have $R_2 \cap R_3 = R_4$.)

Proof. Assume R and S are equivalence relations on set A . We want to prove $R \cap S$ is also an equivalence relation.

First, we show $R \cap S$ is reflexive. Take any element a in A . Since R is an equivalence, then aRa . Since S is an equivalence relation, then aSa . Thus a is related to itself by both R and S , Hence $a(R \cap S)a$. Therefore $R \cap S$ is reflexive.

Second, we show $R \cap S$ is symmetric. Take any $a, b \in A$ such that $a(R \cap S)b$. This means that both aRb and aSb . Since R is symmetric bRa . Since S is symmetric bSa . Thus b is related to a by both R and S , hence $b(R \cap S)a$. Therefore $R \cap S$ is symmetric.

Last, we show $R \cap S$ is transitive. Take any $a, b, c \in A$ and suppose $a(R \cap S)b$ and $b(R \cap S)c$. By definition of intersection it follows aRb , bRc and aSb , bSc . Since R is transitive aRc . Since S is transitive aSc . Thus a is related to c by both R and S , hence $a(R \cap S)c$. Therefore $R \cap S$ is transitive.

We have shown that if R and S are equivalence relations on a set A , then the intersection of R and S is also an equivalence relation. $\square$

12. Prove or disprove: If R and S are two equivalence relations on a set A , then $R \cup S$ is also an equivalence relation on A .

Disproof. The statement is false. For a counterexample, let $A = \{a, b, c\}$ and take the equivalence relations R_2 and R_4 from Exercise 6,

$$R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\} \quad \text{and} \quad S = \{(a, a), (b, b), (c, c), (b, c), (c, b)\}.$$

Then,

$$R \cup S = \{(a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b)\}.$$

We have $(a, b) \in R \cup S$ and $(b, c) \in R \cup S$, but $(a, c) \notin R \cup S$. So $R \cup S$ is not transitive, and hence is not an equivalence relation. $\square$

11.3 Equivalence Classes and Partitions

2. List all the partitions of the set $A = \{a, b, c\}$. Compare your answer to the answer to Exercise 6 of Section 11.2.

$$\{\{a\}, \{b\}, \{c\}\}, \{\{a, b\}, \{c\}\}, \{\{a\}, \{b, c\}\}, \{\{a, c\}, \{b\}\}, \{\{a, b, c\}\}.$$

4. Suppose P is a partition of a set A . Define a relation R on A by declaring xRy if and only if $x, y \in X$ for some $X \in P$. Prove R is an equivalence relation on A . Then prove that P is the set of equivalence classes of R .

To prove the statement above we will use the following consequence: if $X, Y \in P$ and $X \cap Y \neq \emptyset$, then $X = Y$.

Proof. First we show that R is an equivalence relation on A .

Take any $x \in A$. Since the union of the sets in P is A , there is some $X \in P$ with $x \in X$. Then $x, x \in X$, so xRx . Therefore R is reflexive.

Suppose xRy . Then $x, y \in X$ for some $X \in P$. But then $y, x \in X$, so yRx . Therefore R is symmetric.

Suppose xRy and yRz . Then $x, y \in X$ and $y, z \in Y$ for some $X, Y \in P$. Since $y \in X \cap Y$, we have $X \cap Y \neq \emptyset$, so $X = Y$. Thus $x, z \in X$, which means xRz . Therefore R is transitive.

These last three paragraphs show R is an equivalence relation on A . □

Proof. Next we show that P is the set of equivalence classes of R .

First we need to show that if $x \in A$ and $X \in P$ with $x \in X$, then $[x] = X$.

Suppose $y \in [x]$, so xRy . Then $x, y \in Y$ for some $Y \in P$. Since $x \in X \cap Y$, we get $X = Y$, so $y \in X$. This shows $[x] \subseteq X$. Conversely, suppose $y \in X$. Then $x, y \in X$ with $X \in P$, so xRy , which means $y \in [x]$. This shows $X \subseteq [x]$. Hence $[x] = X$.

Now we show that every equivalence class of R belongs to P .

Let $[x]$ be an equivalence class, where $x \in A$. As in the reflexive case, there is some $X \in P$ with $x \in X$, and by the claim $[x] = X \in P$.

Finally, we show that every set in P is an equivalence class of R .

Let $X \in P$. Since X is nonempty, we can choose some $x \in X$. By the claim, $X = [x]$, so X is an equivalence class of R .

Therefore the set of equivalence classes of R is exactly P . □

11.4 The Integers Modulo n

2. Write the addition and multiplication tables for $\mathbb{Z}_3$.

+	[0]	[1]	[2]
[0]	[0]	[1]	[2]
[1]	[1]	[2]	[0]
[2]	[2]	[0]	[1]

·	[0]	[1]	[2]
[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]
[2]	[0]	[2]	[1]

4. Write the addition and multiplication tables for $\mathbb{Z}_6$.

+	[0]	[1]	[2]	[3]	[4]	[5]	·	[0]	[1]	[2]	[3]	[4]	[5]
[0]	[0]	[1]	[2]	[3]	[4]	[5]	[0]	[0]	[0]	[0]	[0]	[0]	[0]
[1]	[1]	[2]	[3]	[4]	[5]	[0]	[1]	[0]	[1]	[2]	[3]	[4]	[5]
[2]	[2]	[3]	[4]	[5]	[0]	[1]	[2]	[0]	[2]	[4]	[0]	[2]	[4]
[3]	[3]	[4]	[5]	[0]	[1]	[2]	[3]	[0]	[3]	[0]	[3]	[0]	[3]
[4]	[4]	[5]	[0]	[1]	[2]	[3]	[4]	[0]	[4]	[2]	[0]	[4]	[2]
[5]	[5]	[0]	[1]	[2]	[3]	[4]	[5]	[0]	[5]	[4]	[3]	[2]	[1]

6. Suppose $[a], [b] \in \mathbb{Z}_6$ and $[a] \cdot [b] = [0]$. Is it necessarily true that either $[a] = [0]$ or $[b] = [0]$?

No. For example, $[2], [3] \in \mathbb{Z}_6$ and $[2] \cdot [3] = [6] = [0]$, yet $[2] \neq [0]$ and $[3] \neq [0]$.
(Compare Exercise 5: in $\mathbb{Z}_5$ this cannot happen, because 5 is prime.)

8. Suppose $[a], [b] \in \mathbb{Z}_n$, and $[a] = [a']$ and $[b] = [b']$. Alice adds $[a]$ and $[b]$ as $[a] + [b] = [a + b]$. Bob adds them as $[a'] + [b'] = [a' + b']$. Show that their answers $[a + b]$ and $[a' + b']$ are the same.

Proof. Since $[a] = [a']$, we have $a \equiv a' \pmod{n}$, so $n \mid (a - a')$ and $a - a' = nk$ for some integer k . Likewise $[b] = [b']$ gives $b - b' = nl$ for some integer l . Adding these equations,

$$(a + b) - (a' + b') = (a - a') + (b - b') = nk + nl = n(k + l).$$

Thus $n \mid ((a + b) - (a' + b'))$, so $a + b \equiv a' + b' \pmod{n}$. By Theorem 11.1 this means $[a + b] = [a' + b']$, so Alice and Bob get the same answer. $\square$